



Fig. 5: Navigation

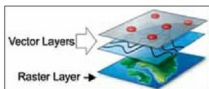


Fig. 6: GIS

ing and controlling a vehicle, craft or a person from same or different places is termed as navigation.

To perform

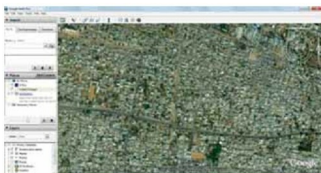


Fig. 7: Google Earth

navigation, GPS data or GPS receiver is not sufficient. We have to use some geographic information system (GIS) that provides a platform on which the GPS data can be seen. GIS is a system by which we can create, visualise, analyse and query geo-spatial data. The geo-spatial data involves geographic information like coordinates, latitude, longitude, geographic location, spatial geometry, map data, etc. The GIS system has various other usages and applications regarding geo-spatial data. There are two types of geo-spatial data:

Raster data. Digital imagery or air photos are the most basic form of

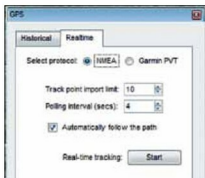


Fig. 8: GPS dialog box in Google Earth



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raster data. Map can also be represented as raster data.

Vector data. Vector data is contained by a layer, super-imposed on the raster layer. Point, line and polygon are three types of vector data. These are maintained by users to create something on the map including village points, rivers, roads, etc, on the raster layer.

GIS can be classified into two categories:

Online. It uses Internet connection to use spatial data. For example, Google Earth and Google Maps.

Offline. First a spatial data is created and then used. For example, ArcGIS and Quantum GIS.

Live navigation using Google Earth

This is an online type of navigation. Internet connection is necessary to view maps and other geo-spatial data. Here we will see the online navigation in a few steps:

1. Run Google Earth. Connect your GPS receiver to appropriate USB/serial port.
2. Go to 'Tools' → GPS.
3. Click on 'Realtime' tab. Select NMEA, don't forget to check 'Automatically follow the path.'
4. Click on 'Start' button. It automatically detects the GPS receiver, processes data and takes you to the position of GPS receiver. ●

The author is a co-founder of Robo India, a leading robotics company